Jerry Lin

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Tech Stack

- Python, PyTorch, Jax, Git, Bash

Work Experience

Postdoctoral Research Associate Sep 2025 - Present
Boston University Faculty of Computing and Data Science (CDS) *Boston, MA*

- Working in the Barnes Group on improving understanding of how AI-weather emulators can be used for scenario planning and simulation of extreme events.

Graduate Research Assistant Sep 2020 - Jul 2025
UC Irvine Earth System Science *Irvine, CA*

- Created an end-to-end pipeline for training and testing hundreds of hybrid physics-AI climate models at a time.
- Core contributor (4th author) to ClimSim, which won the best paper award in the 2023 NeurIPS Datasets and Benchmarks Track.
- Organized the $50,000 2024 LEAP ClimSim Kaggle Competition and submitted a paper investigating how the winning deep learning architectures performed when integrated in a climate model.

Data & Analytics Services (DAS) Intern May 2024 - Sep 2024
Vayuh.ai and Lawrence Berkeley National Laboratory *Berkeley, CA*

- Created a machine learning pipeline for predicting wind power output using real-time HRRR forecasts.

Selected Papers

- Crowdsourcing the Frontier: Advancing Hybrid Physics-ML Climate Simulation via a $50,000 Kaggle Competition (1st author, in review at JAMES)
- Data-driven integration kernels for interpretable nonlocal operator learning (2nd author; published in Environmental Data Science, 2026)
- Navigating the Noise: Bringing Clarity to ML Parameterization Design with O(100) Ensembles (1st author, published in JAMES)
- Stable Machine-Learning Parameterization of Subgrid Processes in a Comprehensive Atmospheric Model Learned From Embedded Convection-Permitting Simulations (4th author, published in JAMES)
- ClimSim-Online: A Large Multi-scale Dataset and Framework for Hybrid ML-physics Climate Emulation (6th author, published in Journal of Machine Learning Research)
- ClimSim: A large multi-scale dataset for hybrid physics-ML climate emulation (4th author, Best Paper in 2023 NeurIPS Datasets and Benchmarks Track)
- Stress-testing the coupled behavior of hybrid physics-machine learning climate simulations on an unseen, warmer climate (1st author, accepted in 2023 NeurIPS Climate Change AI Workshop)
- Climate-invariant Machine Learning (6th author, accepted in Science Advances)

Education

University of California, Irvine Sep 2020 - Jul 2025
PhD, Earth System Science *Irvine, CA*

University of California, Berkeley Aug 2014 - Dec 2018
BA, Applied Mathematics *Berkeley, CA*

Extracurricular Activities & Awards

2024 ML4ESM ICML Workshop organizer

- Conceived of and organized the first Machine Learning for Earth System Modeling (ML4ESM) Workshop at ICML.
- https://leap-stc.github.io/ml4esm-workshop/

2023 Excellence in Leadership Award (UC Irvine Earth System Science Department)